



Answers to your questions about RF, wireless and health

Health

General

5G

Safety & Testing

“Based on our ongoing evaluation of this issue, the totality of the available scientific evidence continues to not support adverse health effects in humans caused by exposures at or under the current radiofrequency energy exposure limits.”

“Recent surveys have indicated that RF exposures from base stations and wireless technologies in publicly accessible areas (including schools and hospitals) are normally thousands of times below international standards . . . From all evidence accumulated so far, no adverse short- or long-term health effects have been shown to occur from the RF signals produced by base stations.”

“... although many studies have examined the potential health effects of non-ionizing radiation from radar, microwave ovens, cell phones, and other sources, there is currently no consistent evidence that non-ionizing radiation increases cancer risk in humans.”

Health

What about reports on social media that there is a relationship between 5G and COVID-19?

Dr. Anthony Fauci has called those stories “garbage.” WHO says, “FACT: 5G mobile networks DO NOT spread COVID-19... Viruses cannot travel on radio waves/mobile networks. COVID-19 is spreading in many countries that do not have 5G mobile networks. COVID-19 is spread through respiratory droplets when an infected person coughs, sneezes or speaks. People can also be infected by touching a contaminated surface and then their eyes, mouth or nose.”

Have wireless devices and networks been shown to cause cancer in humans?

No, according to the international scientific consensus, wireless devices and networks have not been shown to cause cancer in humans. In fact, an analysis of government statistics shows that since the introduction of mobile phones in the 1980s, rates of brain cancer have remained unchanged while the exposure to RF energy from wireless networks has gone up. Many studies have explored whether cellphones cause cancer with the American Cancer Society concluding that “RF waves given off

by cell phones do not have enough energy to damage DNA directly or to heat body tissues. Because of this, it's not clear how cell phones might be able to cause cancer."

Are children at risk?

RF energy at the low levels approved for everyday consumer use has not been shown to pose a health risk to children, teenagers or any demographic group, according to scientific consensus. To cite one example, the Surveillance, Epidemiology and End Results (SEER) program of the National Cancer Institute at the National Institutes of Health **found** no correlation between cancer rates and cellphone use by U.S. children and teenagers between 1992 and 2006, despite a rapid rise in their use of cellphones.

Did a recent National Toxicology Program (NTP) study establish a link between RF energy and cancer in humans?

No, the NTP study did not establish a link between RF energy and cancer in humans. The authors of the study said that their findings did not apply to humans and **that** "the exposure levels and durations in our studies were greater than what people experience." After reviewing the study, the Food and Drug Administration **agreed**, saying that "the existing safety limits for cell phones remain acceptable for protecting the public health."

I saw a research paper that claimed wireless devices and networks might

cause health effects, what does that mean?

As explained in a recent *New York Times* [story](#), the Russian government is promoting discredited claims in an attempt to scare Americans. This includes highlighting activists whose “work appears not in reputable science journals but little-known reports, publications and self-published tracts, at times with copious notes of dubious significance.” It can be difficult for non-scientists to understand whether or not an article is reputable, which is why a recent journal article by researchers at the University of Oxford [explained](#) that “individual studies cannot be relied upon (Schoenfeld & Ioannidis, 2013): the gold standard is a systematic review.” Independent institutions such as the World Health Organization regularly conduct systematic review of all available scientific literature and continue to conclude that radiofrequency energy from wireless devices and networks, including 5G, has not been shown to cause health problems. The evidence consists of thousands of peer-reviewed studies conducted over 70 years.

Can RF energy from cellphones interfere with pacemakers?

Radiofrequency (RF) energy from cellphones can interact with some electronic devices, producing what is known as electromagnetic interference. The Food and Drug Administration (FDA) has established a testing standard for the manufacture of cardiac pacemakers to ensure they are safe from RF. The FDA [advises](#) that anyone looking for an extra precautionary step can hold their cellphone to the ear opposite the side of the body where the pacemaker is implanted, and avoid carrying a turned-on mobile phone in a pocket directly over the pacemaker.

Is RF energy a carcinogen?

No, RF energy has never been classified as a carcinogen. The International Agency for Research on Cancer (IARC) classifies whether substances are known to cause cancer, probably cause cancer or might cause cancer. In 2011 IARC listed RF energy as a possible carcinogen, alongside pickles, coffee and aloe. This classification applies to all forms of RF energy, including broadcast television, broadcast radio, Wi-Fi and Bluetooth signals, and signals from baby monitors, garage door openers and other everyday items. The classification means that IARC believes scientists should continue to research RF energy.

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General

Are cellphones, cell towers, small cells and antennas safe?

Radiofrequency energy from wireless devices and networks, including radiofrequencies used by 5G, have not been shown to cause health problems, according to the international scientific community. To cite one example, the Food and Drug Administration [said](#), “Based on our ongoing evaluation of this issue, the totality of the available scientific evidence continues to not support adverse health effects in humans caused by exposures at or under the current radiofrequency energy exposure limits.”

What is the scientific evidence related to RF and health?

The evidence consists of thousands of peer-reviewed studies conducted over 70 years by independent scientific organizations. The Federal Communications Commission, regulates RF emissions from wireless devices and equipment. In December 2019, the FCC adopted the recommendations of expert organizations that have reviewed the science and [reaffirmed](#) its safety standards on a unanimous and bipartisan basis, saying, "...there is no evidence to support that adverse health effects in humans are caused by exposures at, under, or even in some cases above, the current RF limits. Indeed, no scientific evidence establishes a causal link between wireless device use and cancer or other illnesses."

What do the experts say?

The international scientific consensus is that there are no known health risks from RF energy at the low levels approved for everyday consumer use. This includes the expert opinion of the [World Health Organization](#), the [American Cancer Society](#) and the [Food and Drug Administration](#). "There is no convincing scientific evidence that the weak RF signals from base stations [cell towers] and wireless networks cause adverse health effects," ([World Health Organization](#)).

What is RF energy?

Radiofrequencies (RF), also called radio waves, are a form of energy, like the light you see all around you. They are used to carry broadcast radio and television signals, Wi-Fi and Bluetooth signals, and signals from baby monitors, remote controls, garage door openers and many other everyday items. The general public has been surrounded by sources of

RF energy such as this for more than 100 years, since the introduction of broadcast radio. The current consensus of scientific opinion has not changed—there are no known health risks from RF energy at the low levels approved for everyday consumer use. That is because RF is a low-frequency form of energy, meaning that it is not powerful enough to cause damage to your cells. In fact, the light you see around you every day operates at a much higher frequency than wireless signals, much closer to high-frequency signals that could cause damage, such as X-rays or Gamma Rays. Radiofrequencies have been studied extensively and their scientific properties are well understood. “These waves are the simplest phenomena known in the universe,” (Dr. Eric Swanson, [Testimony 2019](#), Professor of Physics at the University of Pittsburgh).

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5G

Why are people spreading misinformation about 5G online?

A recent [New York Times](#) investigation reveals that the Russian government is “doing its best to stoke the fears of American viewers,” by spreading false information about 5G and health, including “claims that lack scientific support.” As the article makes clear, “plenty of careful science has scrutinized wireless technology for potential health risks. Virtually all the data contradict the dire alarms.” Russia’s efforts appear to be a form of ‘economic warfare’ that draw on discredited research, according to the experts cited by the *New York Times*.

Is 5G safe?

The scientific consensus is that there are no known health risks from all forms of RF energy at the low levels approved for everyday consumer use. The FCC regulates RF emissions, including millimeter waves from 5G devices and equipment, and has adopted the recommendations of expert scientific organizations that have reviewed the science, including dozens of studies focused specifically on millimeter waves, and established safe exposure levels. In December 2019, the FCC [reaffirmed](#)—on a unanimous and bipartisan basis—these safety standards. Typical exposure to 5G devices—such as small cells attached to phone poles or the sides of buildings—is far below the permissible levels and comparable to Bluetooth devices and baby monitors ([New Orleans City Council Hearing](#), 2019). The FCC continues to [monitor](#) the science to ensure that its regulations are protective of public health.

Should federal regulations be updated for 5G?

While 5G networks are new, the FCC regulates RF emissions, including millimeter wave frequencies from 5G devices and equipment, and has adopted the recommendations of expert scientific organizations that have reviewed the science, including dozens of studies focused specifically on millimeter waves, and established safe exposure levels. In December 2019, the FCC [reaffirmed](#)—on a unanimous and bipartisan basis—these safety standards. With the assistance of several federal agencies focused on health issues, the Federal Communications Commission constantly [monitors](#) this research and has not indicated a need to change regulations.

I have heard there is no research on the health effects of 5G technology, is that true?

No it is not true. While millimeter wave frequencies are new to wireless networks, they are extremely well understood by the international scientific community. The IEEE, which the FCC describes as “internationally recognized for [its] expertise in this area,” has assembled a list of dozens and dozens of studies on millimeter wave frequencies. The list of the millimeter wave studies and reviews cited by the IEEE is [here](#). In December 2019, the FCC, which regulates radiofrequency emissions in the United States, adopted the recommendations of expert organizations that have reviewed the science, including from the IEEE, and [reaffirmed](#) that its safety standards “ensure the health and safety of workers and consumers of wireless technology,” and that “no scientific evidence establishes a causal link between wireless device use and cancer or other illnesses.”

What is wireless radiation? Will 5G expose me to more wireless radiation?

Wireless radiation sounds scary, but it’s really just the word scientists use to describe the type of energy that moves in waves, including broadcast radio and television signals and signals from your baby monitor, remote control or cellphone. The scientific consensus is that there are no known health risks from all forms of RF energy at the low levels approved for everyday consumer use. While 5G networks are new, the FCC regulates RF emissions, including millimeter wave frequencies from 5G devices and equipment, and has adopted the recommendations of expert scientific organizations that have reviewed

the science, including dozens of studies focused specifically on millimeter waves, and established safe exposure levels. In December 2019, the FCC [reaffirmed](#)—on a unanimous and bipartisan basis—these safety standards.

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Safety & Testing

Do cellphones and wireless equipment have to meet safety standards?

Yes. Safety standards are [set](#) by the Federal Communications Commission in order to protect public health. In December 2019, the FCC [reaffirmed](#)—on a unanimous and bipartisan basis—these safety standards. The Food and Drug Administration has also [said](#) that “the existing safety limits for cell phones remain acceptable for protecting the public health.” Wireless devices go through a rigorous approval process to ensure they meet guidelines and they operate well under safety limits. These limits are based on recommendations from the scientific community and expert non-government organizations including the National Council on Radiation Protection and Measurements.

Why haven't federal regulations been updated in the past twenty years?

Federal safety regulations have been designed to protect public health and wireless devices operate well below federal safety limits. In fact, the Food and Drug Administration has [said](#) that “the existing safety limits for

cell phones remain acceptable for protecting the public health.” RF energy is the subject of continuous, ongoing research. With the assistance of several federal agencies focused on health issues, the Federal Communications Commission constantly **monitors** this research and has not indicated a need to change regulations. In December 2019, the FCC **reaffirmed**—on a unanimous and bipartisan basis—these safety standards.

What is the standard for testing and assuring the safety of wireless devices?

The primary standard for testing is called the SAR, or Specific Absorption Rate. It is a measure of the rate at which the human body absorbs RF energy. This measurement and safety ranges were developed by internationally recognized expert non-government organizations, including the Institute of Electrical and Electronics Engineers (IEEE), and the National Council on Radiation Protection and Measurements (NCRP). These groups work in consultation with U.S. government agencies, including the Food and Drug Administration (FDA) and Federal Communications Commission (FCC), among others. The **allowable SAR limit** is 1.6 watts per kilogram (W/kg), as averaged over one cubic gram of tissue. This standard, adopted by the FCC in 1996, includes a margin of safety that is 50 times below the level of observed effects from thermal (heat) exposure (Christopher Davis, **Testimony**, 2019). Wireless devices and antennas operate well under FCC thresholds and all wireless devices sold in the U.S. go through a formal FCC approval process to ensure that they comply with the maximum allowable SAR level standard when operating at the device’s highest possible power level.

If RF energy is safe, why does the government have safety limits?

While RF energy is not strong enough to cause cell or DNA damage by ionization of atoms or molecules like X-rays can, high levels of RF exposure could cause heating, which is why governments set exposure limits. Limits for wireless devices are set more than 50 times below the level at which heating caused by RF energy has been shown to have an observed effect (Christopher Davis, [Testimony](#), 2019).

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